PLANNING AND MONITORING & EVALUATION | TECHNICAL ADVISORY REPORT

Tanzania's Agricultural Sector: Progress and Target Readiness

Three AASS rounds assessed against FYDP IV and DIRA 2050

Survey years: 2022/23, 2023/24 and 2024/25 | Report date: 11 November 2026

Decision for the Minister. Protect productivity, not only production volume. Published survey estimates show higher maize and paddy output, but lower output per harvested hectare between 2022/23 and 2024/25. Irrigation use and input adoption improved at the endpoints, with reversals in the middle year. Prioritize functioning irrigation, verified input quality, agronomic extension and a common indicator framework before claiming FYDP IV or DIRA 2050 target attainment.

+43.7%	-21.9%	8.1%	NOT VERIFIED
Maize production 2022/23 to 2024/25	Maize yield per harvested ha same two endpoints	Crop households irrigating 2024/25	Official target attainment definition / evidence gaps

Reading the evidence: maize output rose from 6.97 to 10.02 million tonnes; paddy from 2.44 to 3.95 million tonnes. Recomputed yields fell from 1.81 to 1.41 t/ha for maize and from 2.22 to 1.80 t/ha for paddy. These are descriptive point-estimate changes, not tested causal effects. [S1-S3, Table 3.1]

Scope safeguard. These surveys predate FYDP IV (2026/27-2030/31). The report assesses the starting position and conditional readiness, not results caused by that plan. Only the five supplied PDFs are used as substantive evidence. Published survey estimates are the empirical baseline, not error-free census truth.

1. Methodology and rules for a defensible assessment

Component	Method applied	Interpretation / safeguard
Evidence design	Repeated cross-sectional, published survey estimates. S1-S3 use a PHC 2022 frame, stratified two-stage household sampling and a separate large-farm component.	No household microdata were supplied. Published weighted national estimates are retained; no unweighted average of regions or fictitious panel is constructed.
Comparability	Match geography, holding type, units, denominator and reference period. Main dashboard is Tanzania (Mainland plus Zanzibar). Production and irrigated-area totals include households and large farms; input shares are household-only.	Policy administrative coverage must be reconciled before formal scoring. Agricultural years run October-September; policy fiscal years July-June. Year-ending scenarios approximate timing, not monthly delivery.
Trend arithmetic	For totals: change = 100(x25/x23 - 1); two-interval CAGR = 100[(x25/x23)^(1/2) - 1]. For shares: percentage-point change = p25 - p23. Yield = total tonnes / total harvested ha.	There are three observations but only two annual intervals. Calculations use unrounded source totals; reported percentages retain published precision. Yields are ratios of aggregates, not means of regional yields.
Target screening	First test whether survey and target measure the same thing. Where they do not, use a clearly labelled proxy or mark not assessable. For irrigation, compare flat, additive and compound scenarios; for input shares use a bounded logit sensitivity.	Scenarios are assumptions, not estimated probabilities. No ARIMA, machine learning, high-order trend, causal regression or 25-year statistical forecast is defensible from three annual aggregates.
Uncertainty	Use the published 2024/25 annual production CVs for an approximate single-year normal interval: estimate +/- 1.96 x CV x estimate. Do not infer cross-year significance.	Earlier matched variances, inter-round covariance and ratio-estimator variance are unavailable for this analysis. Scenario ranges are not confidence intervals. Sampling precision does not establish absence of bias.

Selection: purposive monitoring of staple production/productivity, irrigation, inputs and key policy measurement gaps; not an exhaustive sector account or composite performance index. **NR** means no directly comparable published estimate established in the supplied report, not zero. **Proxy** means informative but not the official target indicator.

Methodological anchors within the permitted evidence: S1 PDF pp. 26, 223-228; S2 PDF pp. 22-23; S3 PDF pp. 39-40 and 249-251. These are the survey reports' own design and quality documentation; no external literature is used.

2. Policy target register: what can actually be tested

Policy commitment	Target / deadline in the supplied plan	AASS linkage and admissible verdict
Irrigation [P1:280]	5,000,000 ha by 2030/31; plan baseline 983,446.06 ha (baseline year not stated in this annex row).	AASS measures irrigated planted area, not necessarily unique equipped/command hectares. Conditional pace screen only; neither completion nor official gap is verified.
Quality inputs [P1:166]	At least 45% of farmers access quality agricultural inputs by June 2030.	Seed and fertilizer use are separate household proxies, not certified quality, access, timeliness, or all farmers. No single official attainment rate.
Climate-smart agriculture [P1:166]	40% of actively cultivated agricultural land adopting climate-smart practices by June 2031.	2024/25 reports individual adaptation practices among holdings. A holding share cannot be substituted for an area share or added across overlapping practices.
Extension and finance [P1:166-167]	Officer/expert ratio from 1:20,000 to 1:10,000 by June 2031; formal finance among agricultural entrepreneurs from 15% to 30% by June 2031.	No matched three-round series. Any agricultural loan and information access are not equivalent to these staffing and formal-finance measures.
Sector outcomes [P1:58,279]	By 2030/31: real agricultural GDP growth 10%; food self-sufficiency ratio 130; exports USD 5 billion; post-harvest losses 10% (plan baseline 35%).	AASS crop tonnes are not real sector GDP, food supply-demand balances, customs export values or a comprehensive loss ratio. Do not score these from crop output alone.
Other subsectors [P1:167-169]	By 2031: at least 50% of livestock products with secondary value addition; domestic livestock/fisheries inputs and feeds +50%; fisheries market penetration at least 50%; commercial forest plantations/woodlots at least 1.5m ha; bee-product production +50%.	The selected AASS publications do not establish matched three-year official indicators for these targets. Do not label the whole agricultural sector on track from crop results.
DIRA 2050 [P2:20,68]	A leading food basket in Africa and among the top ten globally; highly productive, competitive, sustainable and technology-enabled agriculture.	AASS supports domestic baseline monitoring. No international comparator/ranking series or numeric 2050 maize yield, fertilizer-use or irrigation target is specified for direct testing.

Important target boundaries: P1 PDF p. 116 describes NAGITA coverage of over 420,000 ha in selected basins. That programme footprint is not the national 5m-ha target, and national AASS hectares cannot establish NAGITA delivery. The climate-smart intervention text also calls for adoption across all agricultural land by 2029, whereas its target row is 40% by 2031: reconcile this internal results-framework tension before setting annual milestones.

3A. Three-year performance matrix: staple output and productivity

Specific issue / indicator	2022/23	2023/24	2024/25	FYDP IV target and reach assessment	DIRA 2050 alignment and reach assessment	Ministerial action to emphasize
Maize production tonnes; all holdings [C1]	6,972,306	7,443,199	10,020,535	**Improving volume.** +43.7% across two intervals; CAGR 19.9%. No matching maize-tonnage target in the cited FYDP agricultural KPI framework. Cannot infer the 10% real-sector-GDP target from maize tonnes.	Supports food-basket potential, but production growth alone cannot verify Africa/global rank or nutrition security. **2050 attainment not estimable.**	Pair production reporting with yield, unit costs, marketed surplus and verified storage losses; emphasize productive output, not planted area alone.
Maize yield tonnes / harvested ha; all holdings [C1]	1.81	1.81	1.41	**Productivity warning.** Ratio-of-totals yield fell 21.9% from 2022/23. No matching numeric crop-yield target established in the cited framework; latest direction is unfavorable.	Lower endpoint yield conflicts directionally with a highly productive sector, but is not a forecast of failure by 2050.	Prioritize soil diagnosis, correct input timing and agronomic extension; monitor harvest-area yield separately from planted-area yield.
Paddy production tonnes; all holdings [C1]	2,444,147	3,134,205	3,952,663	**Improving volume.** +61.7%; CAGR 27.2%. No directly comparable paddy-tonnage target established; not evidence that the food self-sufficiency ratio reaches 130.	Positive food-hub signal, not proof of export competitiveness, food availability per person or top-ten status.	Link functioning irrigation to milling quality, storage and offtake; track paddy separately from milled rice to prevent conversion errors.
Paddy yield tonnes / harvested ha; all holdings [C1]	2.22	2.38	1.80	**Productivity warning.** Endpoint decline 19.1%; the middle-year gain was not sustained. No numerical target-matched reach probability is defensible.	Latest reversal weakens the productivity trajectory; 2050 outcomes remain conditional on sustained change, not three points.	Audit scheme water reliability, seed quality and crop husbandry; compare yields within consistent production systems before diagnosing causes.

C1 source cells: S1 Table 3.1, PDF p. 43; S2 Table 3.1, PDF p. 38; S3 Table 3.1, PDF p. 57. Yields above are recomputed from published totals, not rounded source yield cells. Paddy is not milled rice. Absolute changes and CAGRs describe point estimates, not statistically tested changes.

Planning interpretation. The identity production = harvested area x yield shows that higher aggregate output coincided with substantially more harvested area and lower endpoint yield. It does not show whether land expansion, weather, farm composition, reporting changes or any particular programme caused the results.

3B. Three-year performance matrix: irrigation and inputs

Specific issue / indicator	2022/23	2023/24	2024/25	FYDP IV target and reach assessment	DIRA 2050 alignment and reach assessment	Ministerial action to emphasize
Irrigated planted area ha; all holdings [C2]	605,195	688,757	941,388	**Conditional pace insufficient.** Against the 5m-ha target only if hectares were equivalent: historical CAGR 24.7% vs required 32.1%; compound 2031 scenario 3.54m ha. Official target gap remains unverified.	Direction supports resilience, but no matching numerical 2050 irrigation target is specified. Long-horizon attainability is not estimable.	Reconcile unique equipped, serviceable, actually irrigated and season-summed hectares; fund operation/maintenance and water reliability, not construction alone.
Crop households irrigating % of crop households [C2]	7.4%	5.8%	8.1%	Endpoint +0.7 percentage points; middle-year decline. No matching household-share target. Higher area does not imply broad-based household access.	Only 8.1% irrigating indicates limited household reach; no numerical 2050 household-coverage target permits a reach test.	Track unique beneficiary households, sex of holder, reliable water-days and seasonal use; protect smaller farmers' access.
Improved seed use % of seed-using households [C3]	41.3%	40.8%	50.9%	**Favorable proxy, not attainment.** +9.6 pp. Latest use exceeds the 45% quality-input access threshold numerically, but populations and quality/access definitions differ.	Consistent with technology uptake at endpoints; no direct 2050 adoption threshold. Productivity impact is not established by adoption alone.	Verify certified seed, crop suitability, affordability and on-time access; publish a harmonized quality-input indicator.
Any fertilizer use % of crop households [C4]	44.7%	40.2%	53.5%	**Favorable proxy, not attainment.** +8.8 pp, with a dip in 2023/24. Any organic/inorganic use is not certified quality access or subsidy receipt; cannot score the 45% target directly.	Potential productivity enabler, not proof of higher yield, efficient nutrient use or sustainable agriculture.	Emphasize soil-specific nutrient management and verified quality; track timeliness, quantity and nutrient-use efficiency rather than fertilizer use alone.
Pesticide use % of crop households [C5]	36.8%	37.5%	45.9%	Endpoint +9.1 pp. No comparable use-rate target. Higher use is **not inherently better performance** and does not prove reduced pest damage.	Competitiveness must be balanced with environmental and food-safety objectives; no 2050 reach test from use prevalence.	Emphasize integrated pest management, registered products, safe application and residue monitoring rather than maximizing chemical use.

C2: S1 PDF pp. 105-106; S2 pp. 95-96; S3 pp. 128-130. C3: S1 p. 110; S2 p. 100; S3 p. 135. C4: S1 p. 112; S2 p. 102; S3 pp. 137-138. C5: S1 p. 118; S2 p. 106; S3 pp. 142, 145. Shares are published rounded estimates. Household seed use permits multiple seed types, so categories do not sum to 100%.

3C. Three-year performance matrix: system bottlenecks and missing measures

Specific issue / indicator	2022/23	2023/24	2024/25	FYDP IV target and reach assessment	DIRA 2050 alignment and reach assessment	Ministerial action to emphasize
Climate-smart land share % of actively cultivated land	NR	NR	NR Holding-level practice proxies available [C6]	**Not assessable.** Target 40% by 2031. AASS 2024/25 reports 17.3% of holdings with multiple cropping and 10.9% using climate-adapted varieties, not hectares meeting a CSA definition.	Directly relevant to sustainable, resilient agriculture. No comparable three-year land series or quantitative 2050 CSA target; do not forecast.	Define a CSA practice bundle and area denominator; measure verified hectares and avoid double-counting overlapping practices.
Extension staffing ratio farmers per officer/expert	NR	NR	NR Information-access proxy available [C6]	**Not assessable.** Target 1:10,000 by 2031. 2024/25 crop-management/harvest/processing information access of 7.5% is not a staffing ratio or proof of service quality.	Supports technology-enabled agriculture, but information access is neither digital literacy nor an agronomic outcome.	Reconcile active frontline officers, service areas and unique farmers; combine staffing coverage with timely advisory contacts and quality checks.
Formal finance access % of agricultural entrepreneurs	NR Any agricultural loan: 4.9% of households [C7]	NR	NR	**Not assessable.** Target 30% by 2031, plan baseline 15%. Any loan, including informal lenders/family, is not formal finance; entrepreneurs are not identical to all agricultural households.	Financing is a transformation enabler; no comparable three-year formal-access series supports a 2050 forecast.	Measure formal lender access using a common entrepreneur definition; monitor lending costs, repayment, youth/women access and guarantee performance.
Post-harvest loss ratio official comparable loss measure	NR	NR	NR	**Not assessable.** Target reduction from the plan's 35% baseline to 10% by 2031. Crop use, unsold output, unharvested area and storage capacity cannot substitute for measured post-harvest loss.	Essential to food-hub readiness; production increases alone cannot show retained food, food safety or exportable surplus.	Establish commodity/stage-specific measured losses and a transparent aggregate; target storage, drying, cold chain and handling at identified loss points.

C6: S3 PDF pp. 190-193 (adaptation, Table 6.1 and information, Table 6.2). C7: S1 PDF pp. 125-126, section 5.2/Table 5.1. NR denotes a missing **matched target indicator**, not absence of all related questions or evidence.

No numerical shortfall from unlike denominators. Subtracting 17.3% of holdings from a 40% land-area target, or 4.9% any-loan households from a 30% formal-finance entrepreneur target, would produce an invalid gap. The first ministerial action is to commission harmonized baselines, not publish a spurious delivery rate.

3D. Sector-wide coverage: avoid a crop-only verdict

Specific issue / indicator	2022/23	2023/24	2024/25	FYDP IV target and reach assessment	DIRA 2050 alignment and reach assessment	Ministerial action to emphasize
Livestock modernization / secondary value addition	NR Livestock-keeping households: 60.6%	NR 57.3%	NR 76.9%	**Not assessable.** Target at least 50% of livestock products undergoing secondary value addition by 2031. Household participation is not processing, breed quality, productivity or product value addition.	Modern livestock systems are explicitly envisioned. The participation series [C8] cannot establish modernization or future competitiveness.	Create species/product-level processing and cold-chain baselines; separate livestock population, yield per animal and secondary processed-product share.
Fisheries / aquaculture inputs and market penetration	NR	NR	NR	**Not assessable.** Inputs/feeds +50% and regional/international product-market penetration at least 50% by 2031 require matched baselines and denominators not established here.	Sustainable fishing, aquaculture and cold-chain modernization are goals; farm participation or fish harvest alone cannot establish international market share.	Agree denominator and price/volume basis for market penetration; measure feed quality, cold-chain availability, certification and verified market access.
Commercial forestry / beekeeping	NR	NR	NR	**Not assessable.** Plantation/community woodlots at least 1.5m ha; honey/bee products +50% by 2031. No matched three-round survey series established for these exact targets.	Conservation, sustainable resource use and commercial forestry matter independently of crop growth. No DIRA 2050 reach verdict from these AASS indicators.	Assign accountable forestry/beekeeping indicator owners; reconcile unique managed area and annual bee-product production before projecting.
Real agricultural GDP, exports, employment, food balance and global rank	NR	NR	NR	**Not assessable from these surveys.** 10% real growth, USD 5bn exports, employment share 50%, GDP share 25.6% and self-sufficiency ratio 130 need appropriate sector-wide series.	Food-basket ranking among the global top ten is comparative. Tanzania-only crop surveys provide neither competitor data nor an agreed ranking metric.	Approve a common results dictionary and assign responsibility for each indicator; obtain the appropriate series in a subsequent authorized exercise.

C8: S1 Table 2.1, PDF p. 28; S2 Table 2.1, p. 24; S3 executive summary p. 18 and chapter 2. The household share changed by +16.3 pp between endpoints, but composition, listing and eligibility may also change. This is not a productivity or causal-effect measure. Policy references: P1 pp. 58, 167-169; P2 pp. 20 and 68.

Scope discipline: NR entries do not imply target failure, zero activity, or lack of data outside these five files. They identify what cannot be established within the user's evidence restriction. Related information printed in report introductions is not used to invent a three-year national-accounts or trade series.

4. Conditional target scenarios and statistical precision

Irrigation screening horizon: approximate six annual steps from the agricultural year ending in 2025 to FYDP's 2030/31 endpoint. The table below assumes, solely for screening, that AASS irrigated planted hectares can be treated as the target's hectares. That assumption is unverified.

Irrigation scenario / requirement	Computation	2031 result / interpretation
No further growth	Hold 2024/25 at 941,388 ha.	0.941m ha; 18.8% of the 5m numerical threshold.
Continue mean absolute increment	(941,388 - 605,195) / 2 = 168,096.5 ha/year; add six increments.	1,949,967 ha; 39.0% of threshold; scenario gap 3,050,033 ha.
Continue endpoint compound growth	Historical two-interval CAGR = 24.7%; 941,388 x (941,388 / 605,195)^3.	3,543,143 ha; 70.9% of threshold; scenario gap 1,456,857 ha.
Pace required to reach 5m	(5,000,000 / 941,388)^(1/6) - 1 = 32.1% per year; equivalent straight-line addition 676,435 ha/year.	Required CAGR exceeds historical CAGR by 7.4 percentage points; straight-line annual increment about 4.0 times the historical mean.

Verdict: both growth continuations fall short under the assumed bridge; this is a planning risk signal, not an official infrastructure deficit, probability of failure or causal model. NAGITA's selected-basin footprint must not be tested using national total area.

Quality-input proxy sensitivity: five annual steps to June 2030

Bounded logit screen: L(p) = ln[p/(1-p)], b = [L(p25)-L(p23)]/2, p2030 = 1/{1+exp[-(L(p25)+5b)]}, using fractions. Improved-seed use: flat 50.9%, continued-logit 73.2%; any-fertilizer use: flat 53.5%, continued-logit 73.6%. Both exceed 45% numerically, but neither proves **quality-input access**. The 2023/24 dips show why endpoint momentum is not a reliable probability forecast.

Published 2024/25 precision, not a forecast interval

S3 Annex B (PDF pp. 249-251) gives annual production CVs of maize 5.8% and paddy 7.0%. Applied to the national totals, approximate 95% normal intervals are maize **8.88-11.16m tonnes** and paddy **3.41-4.49m tonnes**, conditional on matching the annex estimate domain. These intervals address single-year sampling precision, not cross-year changes, bias, yield precision or target probability.

2050 boundary: a 25-step extrapolation from two observed intervals is not reported. DIRA's global rank needs international evidence and an operational definition; none is supplied. Reach is therefore **not estimable**, not declared impossible.

5. Ministerial action plan: accountable priorities

The actions below are recommendations, not commitments or cost estimates stated in the source files. Sequence them through the current FYDP IV annual plan and budget process; require costed submissions before financing decisions.

Priority / timing	Ministerial emphasis	Proposed accountable lead / partners	Deliverable and M&E; verification
1. First 90 days	Issue one agricultural results dictionary and reconcile the survey-policy bridge before presenting an on-track/off-track score.	Agriculture planning/M&E; directorate; NPC; NBS/OCGS; relevant sector ministries.	Approved indicator cards: units, geographic scope, numerator, denominator, baseline year, deadline, data owner and uncertainty. Reconcile P1's CSA targets and extension ratio denominator.
2. First 90 days, then quarterly	Audit functionality and actual use of irrigation; protect operations, maintenance and reliable water allocation.	National Irrigation Commission with basin authorities, LGAs, scheme organizations and planning/M&E.;	Scheme register separating unique command/equipped ha, functional ha, season-specific planted-irrigated ha and unique users. Report service reliability and reconcile national AASS with scheme records.
3. Next season / 12 months	Couple quality inputs with soil-informed extension; investigate declining aggregate yields without presuming a cause.	Crop development, research/extension and relevant seed/fertilizer/pesticide regulators; LGAs.	Verified on-time quality-input access using the agreed denominator; crop-cut/area validation where feasible; harvested-area yields, advisory coverage, IPM compliance and input complaints.
4. Within 12 months	Target post-harvest loss and market constraints at commodity/stage level, rather than assuming increased output equals usable surplus.	Marketing/value-chain directorates; storage/processors; livestock/fisheries authorities; LGAs.	Measured loss baselines with harmonized stage definitions; drying/storage/cold-chain utilization, quality rejection and processed-product share. Submit appraised, costed investment priorities.
5. Within 12 months, annual review	Make CSA finance and extension measurable, with inclusion and environmental safeguards.	Planning/M&E;, extension, finance institutions/guarantee managers and sector climate focal points.	Area-based CSA measure without overlap double-counting; formal-finance access and repayment; active officer-to-farmer ratio; sex/youth breakdowns and environmental-risk controls.
6. Every annual sector review	Refresh trends with the next survey, attach precision flags and report subsectors separately.	Planning/M&E; with NBS/OCGS, NPC and livestock/fisheries/forestry/beekeeping indicator owners.	Updated three-plus-round comparable series; domain CVs/SEs, bridge notes, verified infrastructure milestones and separately identified missing measures. Do not invent a sector-wide composite score.

Budget discipline: assess life-cycle cost, O&M funding, procurement readiness and measurable results for each proposal. No monetary benefits, employment effects, production-loss valuation or programme costs are fabricated in this report.

6. Evidence ledger and reproducible calculation inputs

All locators below are **physical PDF page numbers, counted from the first PDF page**. Use these instead of contents-page references, which are not always aligned with printed pagination.

Code / measure	2022/23 source and raw input	2023/24 source and raw input	2024/25 source and raw input
C1. Maize tonnes / harvested ha	S1 p. 43, Table 3.1, Tanzania / all holdings: 6,972,306 / 3,854,979.	S2 p. 38, Table 3.1, Tanzania / all holdings: 7,443,199 / 4,121,487.	S3 p. 57, Table 3.1, Tanzania / all holdings: 10,020,535 / 7,090,868.
C1. Paddy tonnes / harvested ha	S1 p. 43: 2,444,147 / 1,099,946.	S2 p. 38: 3,134,205 / 1,315,876.	S3 p. 57: 3,952,663 / 2,198,600.
C2. Irrigated area / crop households irrigating	S1 pp. 105-106, Tables 4.1-4.2: 605,195 ha all holdings; 656,591 / 8,816,122 crop households; published 7.4%.	S2 pp. 95-96, Tables 4.1-4.2: 688,757 ha all holdings; 510,506 / 8,848,983; published 5.8%.	S3 pp. 128-130, Tables 4.1-4.2: 941,388 ha all holdings; published crop-household share 8.1%.
C3. Improved seed use	S1 p. 110, Table 4.3: 41.3% of 8,679,852 seed-using households.	S2 p. 100, Table 4.3: 40.8% of 8,616,147 seed-using households.	S3 p. 135, Table 4.3: 50.9% of 9,470,337 seed-using households.
C4. Any fertilizer use	S1 p. 112, Table 4.4: 44.7% of 8,816,122 crop households. Do not use the 56.1% inorganic share among fertilizer users as the adoption denominator.	S2 p. 102, Table 4.4: 40.2% of 8,848,983 crop households.	S3 pp. 137-138, Table 4.4: 53.5% of 9,802,016 crop households.
C5. Pesticide use	S1 p. 118, Table 4.9: 36.8% of crop households.	S2 p. 106, Table 4.7: 37.5% of crop households.	S3 p. 145, Table 4.8: 45.9% of crop households; narrative p. 142.
C6-C8. Related proxies, not target substitutes	S1 pp. 125-126: 4.9% any agricultural loan. S1 p. 28, Table 2.1: livestock participation 60.6%.	S2 p. 24, Table 2.1: livestock participation 57.3%. No matched finance, CSA land or staffing indicator used.	S3 pp. 190-193: adaptation and information access; p. 18: livestock participation 76.9%.

Source conflict resolved transparently: S1's key-findings graphic (PDF p. 19) lists 922,852 irrigated ha (4.6%), but section 4.1.2 and Table 4.2 (p. 106) state 605,195 ha (3.5%). This report uses the detailed, component-reconciling Table 4.2, not the graphic. The discrepancy needs publisher clarification; it is not silently averaged.

Production decomposition (descriptive identity): maize harvested area rose 83.9% and yield fell 21.9%, giving output +43.7%; paddy harvested area rose 99.9% and yield fell 19.1%, giving output +61.7%. Use multiplicative factors, not addition of growth percentages. Seasonal/developed hectares are not unique physical land.

7. Data-quality qualifications, source provenance and use

Qualification	Consequence for the Minister's interpretation
Frame, listing and sample changes	S1 sampled 16,224 households in 1,352 EAs; S2 targeted 18,048 and identified 17,759 eligible households in 1,504 EAs; S3 targeted 18,024 and identified 17,843 eligible households in 1,502 EAs. Eligible counts are not response rates. S1 Annex C documents frame/listing, recall/calendar and local-unit conversion risks. Changes in published totals may reflect real change and measurement/composition effects.
Area and input denominators	S3 Table 4.2 explicitly describes annual planted area as developed area summed over seasons. Its total is 23,460,682 ha. S3's 35.2% improved-seed area refers to area planted with seeds (section 4.2.1.2), not that total planted-crop denominator. Household input-use shares are retained; area shares are not spliced into a false common series.
Uncertainty and aggregation	S3 annual maize/paddy national CVs are 5.8%/7.0%, but 5/15 regions respectively have annual production CV above 35% (Annex B). National precision is not regional precision. No standard errors are invented for input shares, yields or trends. No causal programme effect, financial return, formal target success probability or 2050 statistical forecast is claimed.

Authorized evidence inventory. The hashes below identify the exact local PDF versions used; they do not authenticate publication status or eliminate source errors. FYDP annex pages with missing text-to-Unicode mappings were read locally through rendered-page OCR and checked against the target-table layout.

ID	Supplied PDF	Pages	SHA-256 (two lines)
S1	AASS 2022-23 NATIONAL REPORT.pdf	295	87c535946e8187c678b2403bc5279260 863f8df62b83864fe451df00851f19ed
S2	NATIONA_REPORT_AASS2023_2024.pdf	388	96e7fae0b7541ba515d0854ef41c6237 5bd377d01ad7b27fb0525be686e35549
S3	AASS_2024_2025.PDF	471	8203f956b76a873dc0634360805efb70 8486797747299c5588609633aae0c0fb
P1	FYDPIV.pdf	314	50899a5819de9e8c62d67662f722b8c0 1cf624ed334dd8cb0207fcde8c91e307
P2	dira2050.pdf	76	502b904ab1f147bf69e092dab2b084d4 a476a951b5ff4df4567d107887a372dd

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